



Direct Current (DC) Subtransmission Backbone for AI Factories

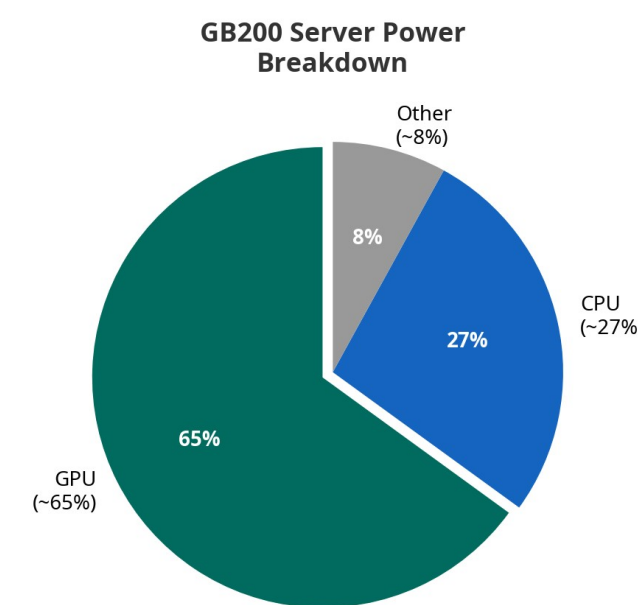
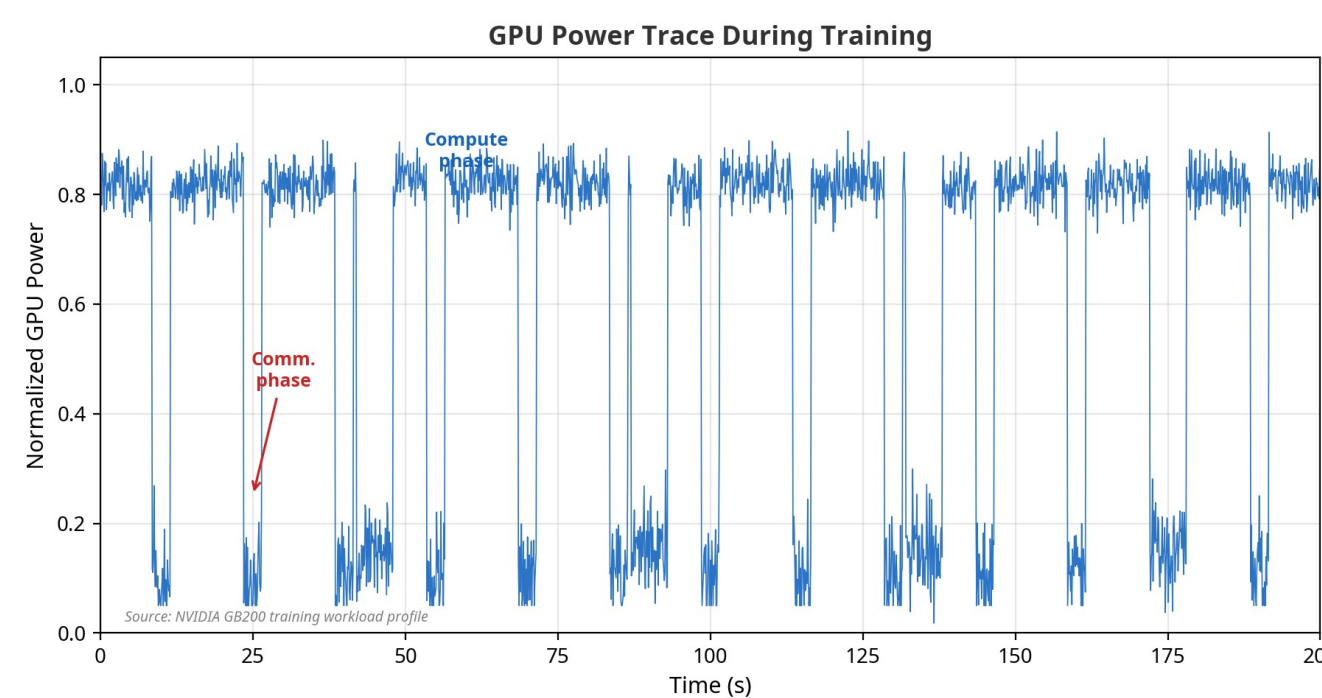
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MOTIVATION

Why Revisit the Power Path Now?

- AI-training loads are changing the problem from a purely steady-state efficiency problem into an architecture + power-dynamics problem.
- The GPU is the dominant power consumer (~65%), and synchronized GPU activity creates periodic power swings visible at the datacenter level.



PROPOSED ARCHITECTURE

Three Scenarios:

1. Traditional AC-centric

Utility MV AC → MV/LV Transformer → UPS (AC → DC → AC) → PDU → Rack PSU (AC → DC) → VRM (DC → DC) → GPU. Six conversion stages. Highest maturity but lowest efficiency.

2. AC-fed SST / 800 Vdc Pod (NVIDIA et al.)

Utility MV AC → Solid-State Transformer (AC-DC) → HF Isolated Link → 800 Vdc Bus.

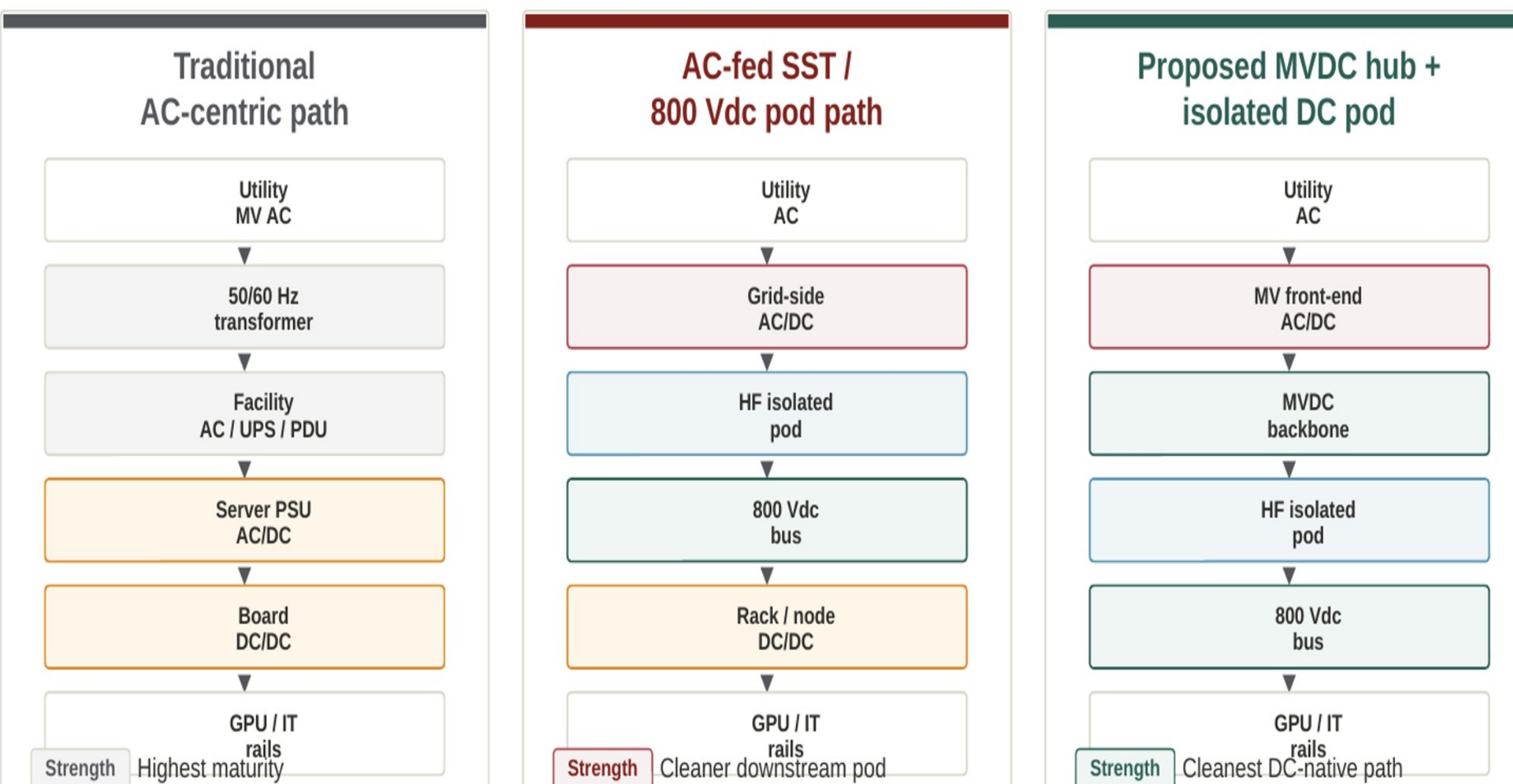
Four conversion stages. Reduces downstream complexity, but AC still enters at the pod front-end each pod must independently handle harmonics and power factor correction.

3. Proposed: MVDC Hub + Isolated DC Pod

Utility MV AC → Centralized MV AC/DC Front-End (PFC + Harmonics) → MVDC Backbone (DC Subtransmission) → HF Isolated DC Pod → 800 Vdc Bus

Three to four conversion stages. Harmonics and PF are handled once at the centralized front-end. The DC backbone natively supports BESS, solar PV, and rack-level buffering.

Target: Utility AC → MV AC/DC → MVDC Hub → Isolated DC Pod → GPU

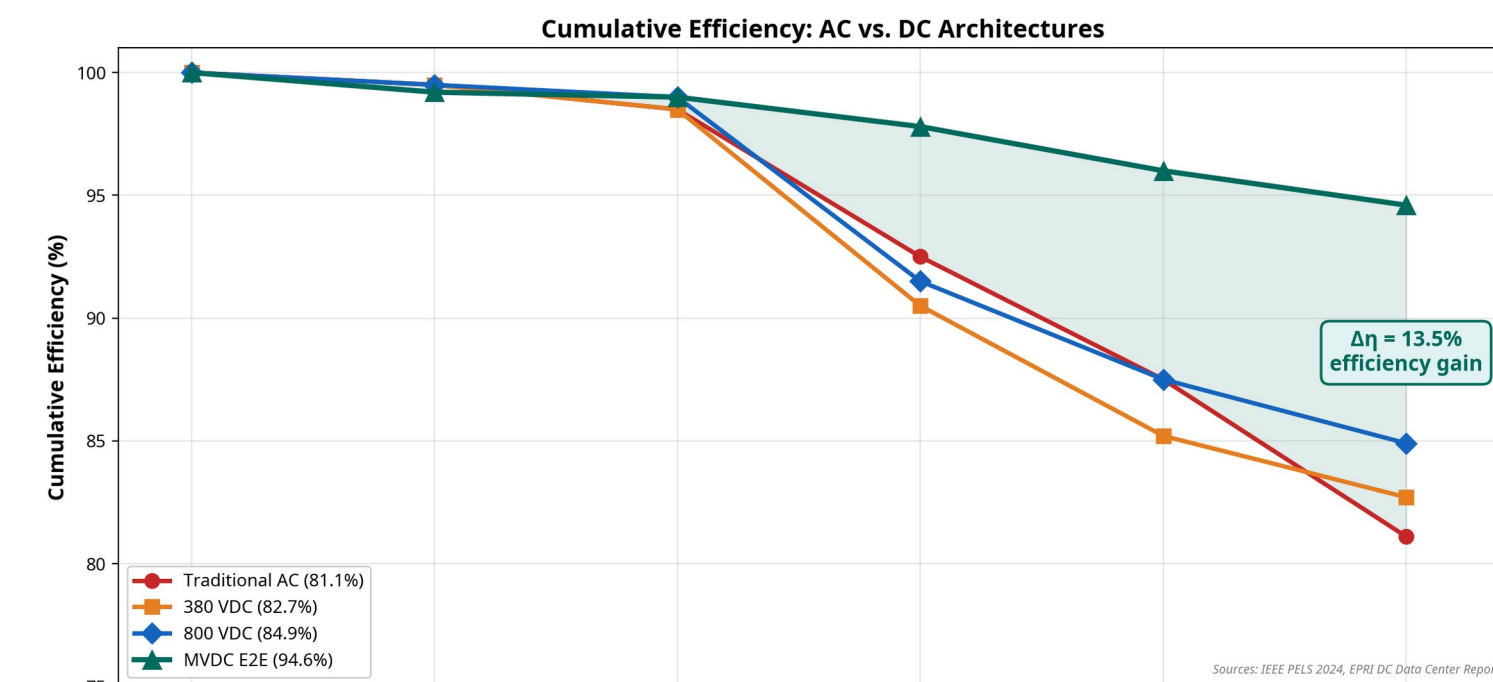


BENEFIT 1: CUMULATIVE EFFICIENCY

The MVDC architecture eliminates redundant conversions.

$\eta_{total} = \prod(\eta_i)$: AC (N=5–6) ≈ 81% vs. DC (N=3–4) ≈ 94.6%

For a 100 MW AI factory: ~13.5 MW saved ≈ \$11.8M/year.



Sources: IEEE PELS 2024; EPRI DC Data Center Report; Rasmussen WP#63

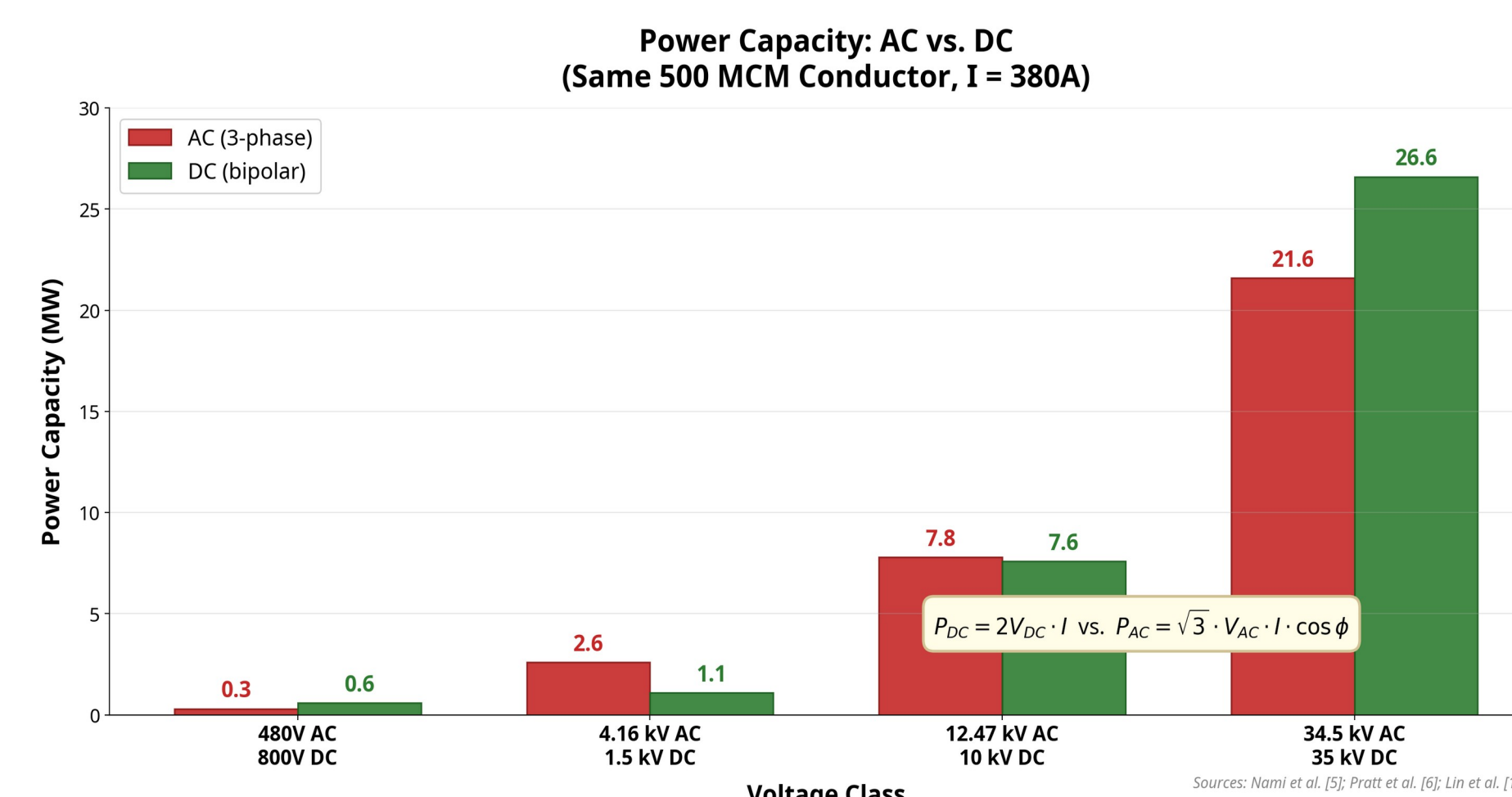
BENEFIT 2: HARMONICS & POWER QUALITY

In AC architectures, every server PSU injects harmonics and requires PFC. Management is distributed across 4–5 points.

With MVDC, harmonic and PF handling become centralized once at the MV AC/DC front-end, instead of

For utilities: AC-side power quality managed only once at the grid interface — simplifying IEEE 519 compliance.

BENEFIT 3: VOLTAGE MGMT & DC-NATIVE PATH



The MVDC backbone provides a cleaner DC-native path for:

- Rack buffering and battery storage (BESS)
- Solar PV integration
- Future modular AI power blocks

All connect directly to the DC bus without AC-DC-AC re-conversion.

DC voltage regulation is inherently simpler: no reactive power, no frequency control, no phase balancing. A single DC voltage setpoint replaces complex multi-variable AC control.

$P_{DC} = 2V \cdot I$ vs. $P_{AC} = \sqrt{3} \cdot V \cdot I \cdot \cos\phi$
DC delivers ~28% more power per conductor at MV level.

ARCHITECTURE COMPARISON

Axis	Traditional AC	AC-fed SST 800 Vdc	MVDC Hub (Proposed)
Facility Interface	AC-native utility + facility protection	AC input still at pod / cluster front end	Centralized MV AC/DC feeding MVDC bus
Pod Complexity	Repeated downstream AC/DC + DC/DC stages	High: PFC + HF link + 800 Vdc output	Simpler: HF link only, AC/DC moves upstream
DC-Native Integration	Extra interfaces for storage / solar / DC	Better local DC, but AC still upstream	Best fit: BESS, solar, rack DC all native
Harmonic / PF Ownership	Distributed across AC facility + servers	Handled locally at each pod interface	Centralized at common MV front end
Maturity	Highest	Prototype to emerging products	Proposed in this work
Main Bottleneck	Repeated conversions, copper, facility sprawl	SiC cost, HFT insulation, capacitor stress	MVDC protection, fault handling

The MVDC-hub proposal is strongest as a system-architecture story

CONCLUSION & FUTURE WORK

Conclusion

The MVDC subtransmission backbone offers a compelling path for next-generation AI factories: fewer conversion stages, centralized power quality management, simpler voltage regulation, and native DC integration for storage and renewables.

Future Work:

- MVDC protection schemes, fault isolation, and safety standards development
- Techno-economic analysis at 100–500 MW scale with hardware-in-the-loop validation
- Optimal buffering placement (GPU, rack, pod, or MVDC hub) and BOM analysis

References

- [1] IEA, "Data Centres and AI," 2024
- [2] IEEE PELS, DC Distribution, 2024
- [3] EPRI, DC Power Report, 2023
- [4] NVIDIA, GB200 Power Architecture, 2024
- [5] Rasmussen, APC WP#63
- [6] Pratt et al., INTELEC 2007
- [7] AILee & Tschudi, IEEE Power & Energy, 2012